



More Sony Exclusive PC ports?

Nixxes, a studio known for porting games to the PC, has been purchased by Sony.
<https://digit.in/jul21-76>



Formula 1 sponsored by Crypto

Formula 1 will be sponsored by crypto.com for \$100 million.
<https://digit.in/jul21-77>

INSTRUMENTATION

Then it comes to the instrumentation, and there are highly specialised AI for just the basslines, rhythms and leads. Sony CSL (Computer Science Laboratories) makes a bunch of tools for artists to programmatically generate rhythmic patterns, as well as specialised software for automatic music processing. Both Amper and Aiva are services that can generate music based on selected parameters such as instrument mix, moods and bpm, allowing creators to match the lyrics generated. The music can also be generated procedurally using more traditional programmatic approaches with software as Max and Ableton. IBM's Watson Beat generates music based on style and emotion. Google's Magenta project includes the Magenta Studio that provides AI based generated music directly as a plugin into Ableton, and the MusicVAE model that allows musicians to generate novel melodies by mixing two pieces of music, as if they were paint. Magenta allows creators to generate songs based on the lyrics in a particular style, so for example, it is totally possible to set a nursery rhyme to a raggaeton beat. Melobytes provides procedurally generated music as a service, and has the innovative capability of converting images into pieces of music. Another model that can generate music based on human domain knowledge inputs is known as FLAGNet. In the back end, a lot of these AI models output MIDI files, which are then piped through to other software solutions downstream.

VOCALS

For the vocals, the simplest approach is to start with a text to speech engine based on the AI generated lyrics (such as <https://ttsmp3.com>), and then process it using music production software such as Ableton. Vocaloid by Yamaha, provides a range of voices and styles, with singing voices that sound genuinely human. Surprisingly singing hides a lot of the flaws of tone deaf text to speech generators,

where even the most refined products in the world such as Amazon's Polly can sound too robotic and unnatural. AI generated vocals are hauntingly beautiful, because you actually get hit by the emotions from a supposedly unfeeling machine. There are also neural networks trained to replicate the voice of particular artists, and one of the most widely used of these is Jukebox by OpenAI that produces tracks directly as raw audio. There are songs generated by Jukebox that replicate the voices of legendary singers such as Elvis Presley and Frank Sinatra. Jukebox is not yet ready for being integrated into professional workflows simply because it takes up to nine hours to create a minute of music. The project known as Lost Tapes of the 27 Club, uses AI generated music in a poignant way,

tion pressure" of human selection. There are some incredibly complex things that artists can do with the music, particularly making it tuned to the situation and the time. Brian Eno's Reflection is an app that plays back ambient music depending on the time of the day, which the artist compared to the changing sounds of the same river. The app Harmonium produces sounds directed to each ear to simulate the calming sounds of ocean waves, based on the local tides, along with the ambient sound for that time (think birds in the morning and insects in the evening). There is a lot of experimentation around generative ambient music for therapeutic purposes, particularly when it comes to delivering binaural beats or coloured noises to match with natural circadian rhythms and brain activity.



Magenta Studio plugins for Ableton Live

by creating tracks from artists that died at the age of 27, including Kurt Cobain, Jimi Hendrix and Amy Winehouse. An album has been released on Spotify by the project, featuring songs that these artists never came around to recording. The initiative is an effort to raise awareness about mental health issues that are particularly prevalent in the music industry.

NOVEL TECHNIQUES

Then there are the artists who create entirely novel production pipelines, such as Holly Herndon who used a custom AI known as SPAWN to harmonise with their own voice, and Karine Miras who created an evolutionary algorithm known as AJ that uses the principles of Darwinian evolution to compose progressively better tracks based on the "popula-

THE BEST AI MUSICIAN

When it comes to generating the lyrics, the most ubiquitous tools are GPT-2 and GPT-3. For the music, we have to go with Google's Magenta, in terms of sheer technical capabilities and the number of different potential applications. When choosing just the best voice, while the AI does cheat a bit by imitating existing or once existing singers who were at the top of their fields, Jukebox does seem to take the cake. However, there is a contest that has been designed to answer this exact question, and every year participants generate tracks. Interested readers can vote on all the tracks, read up on the specific techniques used for each composition, and vote on their favorite pieces of music, over at <https://www.aisongcontest.com>.